

EDUCATION

Master of Science, Software Engineering*University of Engineering and Technology (UET), Taxila, Pakistan*

Oct 2018 – Dec 2022

CGPA 3.33/4.00

- Supervisor: [Prof. Ali Javed](#)
- Thesis: "Deep Learning Model for the detection and classification of Diabetic Retinopathy"
- Major courses: Algorithm Design & Analysis, Advanced Software Technologies, Deep Learning

Bachelor of Science, Software Engineering*Department of Software Engineering, University of Engineering & Technology (UET), Taxila, Pakistan*

Oct 2014 – July 2018

CGPA 3.50/4.00

- Supervisor: [Prof. Syed Muhammad Anwar](#)
- Final Year Project: "Context-Aware Smart Car App for Personal Assistance"
- Major courses: Artificial Intelligence, Web Development, Software Testing, Software Quality

RESEARCH INTERESTS

- Brain-computer interfaces (BCIs) and neural signal processing using EEG and ECoG
- Decoding inner and imagined speech from brain activity using deep learning models
- Deep learning for neural decoding and assistive brain-based communication systems

PUBLICATIONS

Deep Learning Model for the detection and classification of Diabetic Retinopathy ([Paper](#))

(ICTAPP-2023)

*Samra Nawazish, Ali Javed, Tahira Nazir*Studying the Invasion of Drones in Indigenous areas using Machine Learning Techniques ([Paper](#))

(UCP Journal- 2024)

*Samra Nawazish, Amin Ullah, Jawad Hassan*Performance Evaluation of MANET Routing Protocols Based on Size and Speed Parameters ([Paper](#))

(UCP Journal-2024)

*Amin Ullah Samra Nawazish, Jawad Hassan*Image Fusion using Wavelet Transformation and XGboost Algorithm ([Paper](#))

(CMC-2024)

*Faten S. Alamri, Dr Shahid Naseem, Tariq Mahmood, Samra Nawazish, Tanzila Saba*Towards sustainable smart cities: Applications, Challenges and Future Directions ([Paper](#))

(IJDSA-2025)

*Amin Ullah, Saddam Hussain, Samra Nawazish, Irfan Ullah, Zafar Ali*Enhancing Dental Caries Identification with Deep Learning: A study of Convolutional Networks and Transfer Learning Approaches ([Paper](#))

(ssrn 4945802-2024)

Fayqa Manan, Muhammad Adeel, Samra Nawazish, Amin Ullah, Amjad Rehman

PROFESSIONAL EXPERIENCE

Researcher

April 2025 – Dec 2025

*Intelligent Machines Lab, Information Technology University (ITU), Lahore, Pakistan*Advisor: [Dr. Waqas Sultani](#)

- Working on real-time Microscopic Images and Video Enhancement and Stitching tool
- Trained [Segformer](#) on Hemato dataset to accurately generate segmentation masks for Blood Cancer dataset Leukemia
- Using [Transformers](#), developed a language translator that translates given text in English to Urdu
- Implemented [DRA-Net](#) for semantic cell segmentation on the BCCD dataset in Whole Slide Images
- Applied [ResNet-50](#) to classify flowers across seven categories

Teaching Assistant

Aug 2025 – Dec 2025

Instructor: [Dr. Mohsen Ali](#) Postgraduate Course Computer Vision CV525, ITU, Lahore, Pakistan**Graduate Research Assistant**

Aug 2021 – Jan 2023

Department of Software Engineering, University of Engg. and Tech., Taxila, Pakistan

Research topic: Deep Neural Networks

- Explored Deep Learning models and transfer learning algorithms for the detection and classification of Diabetic Retinopathy

Lecturer

July 2021 – Dec 2024

Swedish College of Engineering & Technology, Wah Cantt, Pakistan

- Taught Courses: Programming Fundamentals, Object Oriented Programming, Introduction to Computers, Software Testing, Artificial Intelligence

Junior Software Developer

June 2019 – June 2021

Heavy Industries Taxila (HIT), Taxila Cantt, Pakistan

- I design, build, and maintain front-end, back-end, and database systems to deliver seamless, user-friendly applications

SUMMER SCHOOLS / INTERNSHIPS / WORKSHOPS / CONFERENCES

- Attended RISE-MICCAI Summer School (virtual) from July 14–18, 2025
- Attended RISE-MICCAI Summer School-Local Activity (virtual) from July 14-17, 2025 organized by MBZUAI
- Attended 2024 IEEE / OES China Ocean Acoustic Conference on May 29-31, 2024 in Harbin, China
- Attended China-Pakistan Marine Information Workshop and Training Course organized at Harbin Engineering University, Harbin, China from June 01-14, 2024
- Attended International conference on computing technologies, tools, and applications ICTAPP-23 on May 09-11, 2023, in Peshawar, Pakistan

HONOURS & AWARDS

- Fully funded Masters scholarship holder from UET Taxila, Pakistan
- Undergraduate merit-based scholarship holder from Federal Government Employee Quota
- Awarded Prime Minister Laptop Scheme Gift 2016
- Academic board topper excellence award in ICS (FBISE Islamabad) 2014
- Academic board topper excellence award in SSC (FBISE Islamabad) 2012

ACADEMIC SERVICES

- Volunteer reviewer of Information fusion (in the area of medical imaging and IoMT)
- Designed Course Learning Outcomes and Program Learning Outcomes at Swedish College

LANGUAGES

English, Urdu, Punjabi

OTHER INTERESTS

- Exploring culture, Gardening, Cooking, Watching Tech videos

REFERENCES

- Dr. Ali Javed, Associate Professor, University of Engineering & Technology, Taxila Pakistan
Email: ali.javed@uettaxila.edu.pk Contact : +92334 5174516
- Dr. Amin Ullah, Associate Professor, Bahria University Lahore Campus, Lahore Pakistan
Email: aminullah.bulc@bahria.edu.pk Contact : +92333 9557815
- Dr. Tariq Mahmood, Post-doctorate Research Fellow, AIDA Lab, Prince Sultan University, Saudi Arabia
Email: tmahmood@psu.edu.sa Contact : +966532042334